



American International University-Bangladesh (AIUB)
Department of Computer Science
Faculty of Science & Technology (FST)

PROJECT TITLE

BuaBondhu - Connecting Reliable Domestic Helpers with Households in Need

Submitted By

Semester: Summer_24_25		Section:	Group Number:	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
	Mst. Sumaiya Siddika	23-51348-1		
	Hossain Mohammad Sami	23-51354-1		
	MD ZARIF IQBAL	23-51384-1		
	D.M. Sadman Sakib.	23-51373-1		
	Mominul Islam	23-51372-1		

The project will be Evaluated for the following Course Outcomes

CO3: <i>Select</i> appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation regarding process model selection	[5Marks]
	Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]
Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report		[5Marks]
CO4: <i>Develop</i> project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
Develop the project plan, its components of the proposed software products		[5Marks]

Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: Mst. Sumaiya Siddika Student ID: 23-51348-1 Contribution in Percentage (20%): <u>Contribution in the Project:</u> - Manages the whole project, tracks progress, and communicates with the team. - Designs the look and feel of the app so users can use it easily. - Gets job offers, creates a profile, and shares availability. _____ Signature of the Student
Student Name: Hossain Mohammad Sami Student ID: 23-51354-1 Contribution in Percentage (20%): <u>Contribution in the Project:</u> - Writes the code for the app (frontend and backend). - Designs the look and feel of the app so users can use it easily. - Helps domestic workers create profiles and verify their documents. _____ Signature of the Student
Student Name: MD ZARIF IQBAL Student ID: 23-51384-1 Contribution in Percentage (20%): <u>Contribution in the Project:</u> - Manages helper profiles, bookings, user info, and keeps data safe. - Designs the look and feel of the app so users can use it easily. _____ Signature of the Student

Student Name: **D.M. Sadman Sakib.**

Student ID: **23-51373-1**

Contribution in Percentage (**20%**):

Contribution in the Project: (**20%**)

- Designs the look and feel of the app so users can use it easily.
- Builds the mobile app version for Android/iOS.
- Gives feedback on the app and uses it to find and hire helpers.

Signature of the Student

Student Name: **Mominul Islam**

Student ID: **23-51372-1**

Contribution in Percentage (**20%**):

Contribution in the Project:

- Builds the mobile app version for Android/iOS.
- Designs the look and feel of the app so users can use it easily.
- Tests the app to find and fix bugs before it is released.

Signature of the Student

1.1 Background and Problem Domain

In many households, especially in cities, domestic helpers (commonly known as "bua") play an important role in daily life — from cleaning and cooking to taking care of children or the elderly. But the process of hiring them is often informal, risky, and unorganized. Most employers rely on personal references or word-of-mouth to find helpers. This can lead to: Mistrust, because helpers are not verified. Mismatched expectations between employers and workers. No safety or legal protection in case of problems. On the other hand, many domestic helpers — especially women from underprivileged and less-educated backgrounds — face big challenges: They don't know how to use apps or websites to find work. They don't have proper documents like NID for identity proof. They often travel long distances, spending extra time and money. They are at risk of being underpaid, harassed, or even exploited. This situation creates problems for both parties: employers cannot find trusted helpers easily, and helpers struggle to get safe and fair jobs. BuaBondhu aims to solve this by creating a smart, tech-based platform that connects households with verified domestic helpers. The unique part of this project is the role of Agencies: Helpers who are not digitally skilled will get help from Agencies to create their profiles. These Agencies will verify their identity, location, and skills. When a family needs a bua, the app (through the Agencies) will suggest 3 suitable helpers based on the family's requirements. From those 3, the employer can choose the one that matches their needs best. This system helps build trust, ensures transparency, and creates safe job opportunities — especially for women who are often left out of digital platforms.

Root Cause of the Problem

For **Customer**, the root causes include:

- No verified list of helpers to choose from.
- Helpers may not match the household's needs.
- No way to check past performance or ratings.
- Takes time to find the right helper nearby.

For **domestic helpers**, the root causes are:

- Most helpers don't know how to use apps or websites
- Many don't have ID cards or documents.
- They often travel far to find jobs.
- They can be underpaid or treated badly.
- They don't get support to find safe and fair jobs.

Importance of Solving this Problem

Solving this problem is critical for several reasons:

- **Societal Impact:** It reduces unemployment among domestic helpers, promotes fair wages, and improves their livelihoods.
 - **Economic Impact:** Streamlines hiring, saving time and resources for households while creating job opportunities for helpers.
 - **Safety and Trust:** Verified profiles and reviews minimize risks for employers and ensure safer working conditions for helpers.
 - **Convenience:** A centralized platform with real-time updates and location-based search enhances accessibility and efficiency.
- This issue affects household efficiency, safety, and the well-being of both employers and helpers, making it a pressing societal challenge that requires a technology-driven solution.
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1.2 Project Objectives

- Simplify the process of finding and hiring domestic helpers.
 - Ensure safety and trust through verified profiles and reviews.
 - Provide fair employment opportunities for helpers.
 - Improve service quality with a transparent rating system.
 - Agencies will help helpers create their profiles.
 - Match helpers with families based on what they need.
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Proposed Solution

- Mobile and web-based application for hiring domestic helpers.
 - Employers: Households can search, verify, and book helpers based on their needs.
 - Helpers: Domestic workers can create profiles, showcase skills, and find employment opportunities.
 - Real-Time Matching: Location-based search and availability filters.
 - Secure Communication: In-app chat with hidden contact details for privacy.
 - Verification System: National ID (NID) checks, police clearance, and employer references.
 - Agencies will support helpers to get listed on the app.
 - The app shows 3 helpers who match the family's needs.
-

Why This Solution is Appropriate

The solution is particularly appropriate because:

- It addresses the core issues of trust, transparency, and efficiency by leveraging technology to formalize an unregulated sector.
- The app is simple and easy to use. Agencies will help helpers who can't use technology.

- The use of verification systems (NID, police clearance) and reviews ensures safety and reliability, critical for both employers and helpers.
- Location-based search and real-time matching reduce inefficiencies, such as long commutes, and improve accessibility.
- The platform empowers domestic helpers by providing access to fair job opportunities and verified work histories, addressing exploitation risks.
- It is feasible as it uses existing technologies (mobile apps, web platforms, GPS, encryption) tailored to the local context of Bangladesh.

Feasibility to Meet Business Objectives

The solution is feasible because:

- There is a high demand for domestic helpers, especially in urban areas.
 - Agencies are already engaged with underprivileged women and can support their onboarding to the platform.
 - The app can be developed using existing, cost-effective technology tools.
 - The platform has strong potential to expand to other cities and offer additional household services over time.
 - Revenue generation is feasible through future premium features and value-added services.
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Basic Functionalities

- Map: Locate available helpers nearby.
 - Direct Contact: Connect with helpers via secure in-app chat.
 - User Profiles: Save preferences, view booking history, and manage hires.
 - Notification System: Receive alerts for bookings, cancellations, and new helper availability.
 - Helper Profiles: View verified details like age, photo, services, and availability.
 - Booking System: Schedule helpers for specific dates and times easily.
 - Review & Ratings: Employers can leave feedback to build trust.
 - Emergency Replacement: Instantly get alternative helper suggestions if one cancels.
 - Agencies Dashboard: Agencies can assist helpers in registration and track their job matches.
 - Security: OTP-based login and optional police verification for added safety.
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Target Users and Benefits

- Households/Employers: Find reliable helpers quickly and safely.

- Domestic Helpers: Get job help from Agencies, Don't need to use apps alone and Better pay and safe work.
 - Agencies: Expand their reach and ensure better placement rates.
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Key Features

1) Helper Profiles:

- Name, age, contact details (verified).
- Services offered (cleaning, cooking, babysitting, etc.).
- Work experience, languages spoken, and availability.
- Preferred locations and profile photo.

2) Rating and Review System:

- Employers can rate helpers based on performance.
- Builds trust and helps others make informed decisions.

3) Location-Based Search:

- Filters for service type, availability, and language.

4) Booking System:

- Schedule hires for specific dates and times.
- Notifications and reminders for bookings.
- Security Measures:
 - OTP-based login and end-to-end encryption.
 - Optional police clearance and reference checks.

5) Agency Support Integration:

- Partner agencies assist helpers in creating and updating their profiles.
- Agencies verify helper identity, skills, and work location.
- Agencies access a dashboard to track job placements and support helpers.
- Helps workers who are not tech-savvy get digital access and opportunities.

6) Emergency Replacement:

- Backup suggestions if a helper is unavailable.

7) Future Expansion:

- Include additional services like nannies, nurses, cooks, or handymen.
 - Expand to office and commercial cleaning staff.
 - Introduce subscription models for long-term hires.
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Contribution to Scientific Results

BuaBondhu advances technology and science by:

- Creating a scalable model for informal labor markets, applicable to developing economies.
- Demonstrating digital platforms' impact on reducing unemployment and enhancing service quality.
- Providing a case study on integrating verification systems (e.g., NID, police clearance) with mobile technology.
- Offering insights into user behavior for location-based platforms, contributing to human-computer interaction and labor economics research.
Outcomes will be documented via user feedback, hiring success metrics, and case studies, enriching academic and industry knowledge.
- Highlights the impact of agency-based support in helping non-digital workers use tech platforms

Literature Review

Several studies have explored digital platforms for labor markets, particularly in informal sectors:

- **Studies on Informal Labor Markets:** Research by the International Labour Organization (ILO) highlights the challenges of informal domestic work, including exploitation, lack of documentation, and poor working conditions. These studies emphasize the need for formalization through technology (ILO, 2020).
- **Gig Economy Platforms:** Studies on platforms like Uber and TaskRabbit (e.g., Sundararajan, 2016) discuss how digital platforms improve efficiency in labor markets but often lack mechanisms for trust and verification in informal sectors like domestic work.
- **Local Context:** Research on Bangladesh's labor market (e.g., Hossain et al., 2018) identifies the lack of centralized platforms for domestic workers, with most hiring occurring through informal networks.

Existing Software Solutions

Current solutions in the domestic helper hiring space include:

- **Local Platforms:** Sheba.xy and Bikroy.com offer limited listings without robust verification or real-time features.
- **Global Platforms:** UrbanClap and Helpling lack Bangladesh-specific features like language support or local verification.
- **Informal Networks:** Word-of-mouth and agencies dominate but lack transparency and scalability.
- **BdTask Housekeeper App:** Not popular, lacks safety features.

How BuaBondhu Extends Existing Solutions

BuaBondhu improves by:

- **Localized Design:** Supports local languages and NID integration.
- **Verification and Safety:** Emphasizes NID checks, police clearance, and reviews.
- **Real-Time Features:** Offers location-based search and emergency replacements.
- **Empowering Helpers Through Agencies:** Agencies assist helpers in profile creation, digital onboarding, and maintaining verified job histories.
- **Scalability:** Plans for expanded services and subscription models ensure long-term relevance.

These enhancements deliver a user-centric solution tailored to Bangladesh's domestic helper market using modern technology.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model

To build the BuaBondhu app, we will use the Agile Model.

Why Agile is the best for this project:

- BuaBondhu is a mobile and web app that needs regular updates and user feedback.
- It has many features like real-time location, profile creation, chat, and booking system that need to be tested and improved step-by-step.
- Reduces risk by testing small parts regularly, fixing bugs early.
- It's easy to change or improve features after getting feedback from users (like employers, helpers, and agencies).
- Daily team collaboration ensures faster problem-solving and smooth progress.
- **Scalable** – new services (like nurses or handymen) can be added without breaking the system.
- Continuous testing improves app quality and reduces last-minute bugs.
- Since this is a new kind of app for Bangladesh, Agile helps in adjusting plans easily if we find any issues.

Why not other models:

- **Waterfall** is too strict; once you finish a step, you can't go back.
- **Spiral** is complex and best for high-risk or very large projects.
- **V-Model** is good for testing but not flexible for changes.
- **Scrum** since its strict roles, ceremonies, and higher cost are not practical for our small project team.

So, **Agile** is the best because it gives flexibility, supports fast changes, and helps deliver the app in small working parts.

2.2 Project Role Identification and Responsibilities

Here are the main roles in this project:

Name	Role	Responsibility
Mst. Sumaiya Siddika	Project Manager	- Manages the whole project, tracks progress, and communicates with the team. - Designs the look and feel of the app so users can use it easily. - Gets job offers, creates a profile, and shares availability.
	UI/UX Designer	
	Domestic Helper (User)	
Hossain Mohammad Sami	Software Developer	- Writes the code for the app (frontend and backend). - Designs the look and feel of the app so users can use it easily. - Helps domestic workers create profiles and verify their documents.
	UI/UX Designer	
	Agency Representative	
MD ZARIF IQBAL	Database Engineer	- Manages helper profiles, bookings, user info, and keeps data safe. - Designs the look and feel of the app so users can use it easily.
	UI/UX Designer	
D.M. Sadman Sakib	UI/UX Designer	- Designs the look and feel of the app so users can use it easily. - Builds the mobile app version for Android/iOS. - Gives feedback on the app and uses it to find and hire helpers.
	Mobile App Developer	
	Client/Employer (User)	
Mominul Islam	Mobile App Developer	- Builds the mobile app version for Android/iOS. - Designs the look and feel of the app so users can use it easily. - Tests the app to find and fix bugs before it is released.
	UI/UX Designer	
	QA Tester	

3. Specific Requirements

3.1 Functional Requirements

3.1.1 Employer (Household) Requirements

1.0	Home Page
1.1	Registration
1.1.1	Taking full name
1.1.2	Taking username
1.1.3	Taking email/phone number
1.1.4	Taking password
1.1.5	Submitting registration form
1.1.6	Sending OTP for verification
1.2	Login
1.2.1	Taking username
1.2.2	Taking password
1.2.3	Forgot password (OTP verification)

1.3	Profile Management
1.3.1	View personal information
1.3.2	Edit and update household details

1.4	Helper Search & Matching
1.4.1	Search by location
1.4.2	Search by service type (cleaning, cooking, babysitting, etc.)
1.4.3	Search by availability
1.4.4	View top 3 matched helpers suggested by the system

1.5	Booking System
1.5.1	Book helper for a specific date/time
1.5.2	Cancel or reschedule booking
1.5.3	Receive booking notifications and reminders

1.6	Communication
1.6.1	In-app chat with helpers (contact hidden)

1.7	Ratings & Reviews
1.7.1	Leave feedback for a helper after booking
1.7.2	View ratings given by other employers

3.1.2 Helper (Domestic Worker) Requirements

2.0	Profile Management
2.1.1	Registration (with agency help if needed)
2.1.2	Uploading ID documents (NID, police clearance, references)
2.1.3	Selecting skills (cleaning, cooking, babysitting, etc.)
2.1.4	2.1.4 Setting preferred location and availability
2.2	Login
2.2.1	Taking username
2.2.2	Taking username

2.3	Job Management
2.3.1	View booking requests
2.3.2	Accept or reject bookings
2.3.3	Update availability status

2.4	Communication
2.4.1	Chat securely with employers

2.5 Ratings & Reviews

2.5	Ratings & Reviews
2.5.1	View feedback from employers

3.1.3 Agency Requirements

3.0	Agency Dashboard
3.1	Register/login as agency
3.2	Add new helper profiles
3.3	Verify helper documents (NID, clearance, references)
3.4	Track job placements for helpers
3.5	Update helper availability/status

3.1.4 System Features

4.0 Matching & Security

4.0	Matching & Security
4.1	Suggest 3 best helpers for each employer request
4.2	Emergency replacement if booked helper cancels

4.3	OTP-based login
4.4	End-to-end encrypted chat

3.2 Non-Functional Requirements

5.0 Performance

5.0	Performance
5.1	The app must support 5,000 users without performance issues.
5.2	Search results should be returned in less than 2 seconds.

5.3 Security

5.3	Security
5.3.1	Data must be encrypted.
5.3.2	Login with OTP and password protection.

5.4	Usability
5.4.1	App must support Bangla and English.
5.4.2	Interface must be simple for less tech-savvy users.

5.5 Availability

5.5	Availability
5.5.1	The system must be available 24/7.

6. Effort Estimation:

6.1 COCOMO : We Select Organic Software Project type because if SLOC (2KLOC to 50KLOC) than it's a Organic Type. Assume our project SLOC is 15k

$$\text{Effort} = \text{PM} = \text{Coefficient}_{\langle \text{Effort Factor} \rangle} * (\text{SLOC}/1000)^P$$

$$= 2.4 * (15000/1000)^{1.05}$$

$$= 41.3$$

$$\text{Development time} = \text{DM} = 2.50 * (\text{PM})^T$$

$$= 10.1 \text{ months}$$

$$\text{Required number of people} = \text{ST} = \text{PM}/\text{DM}$$

$$= 4.1 \sim 5$$

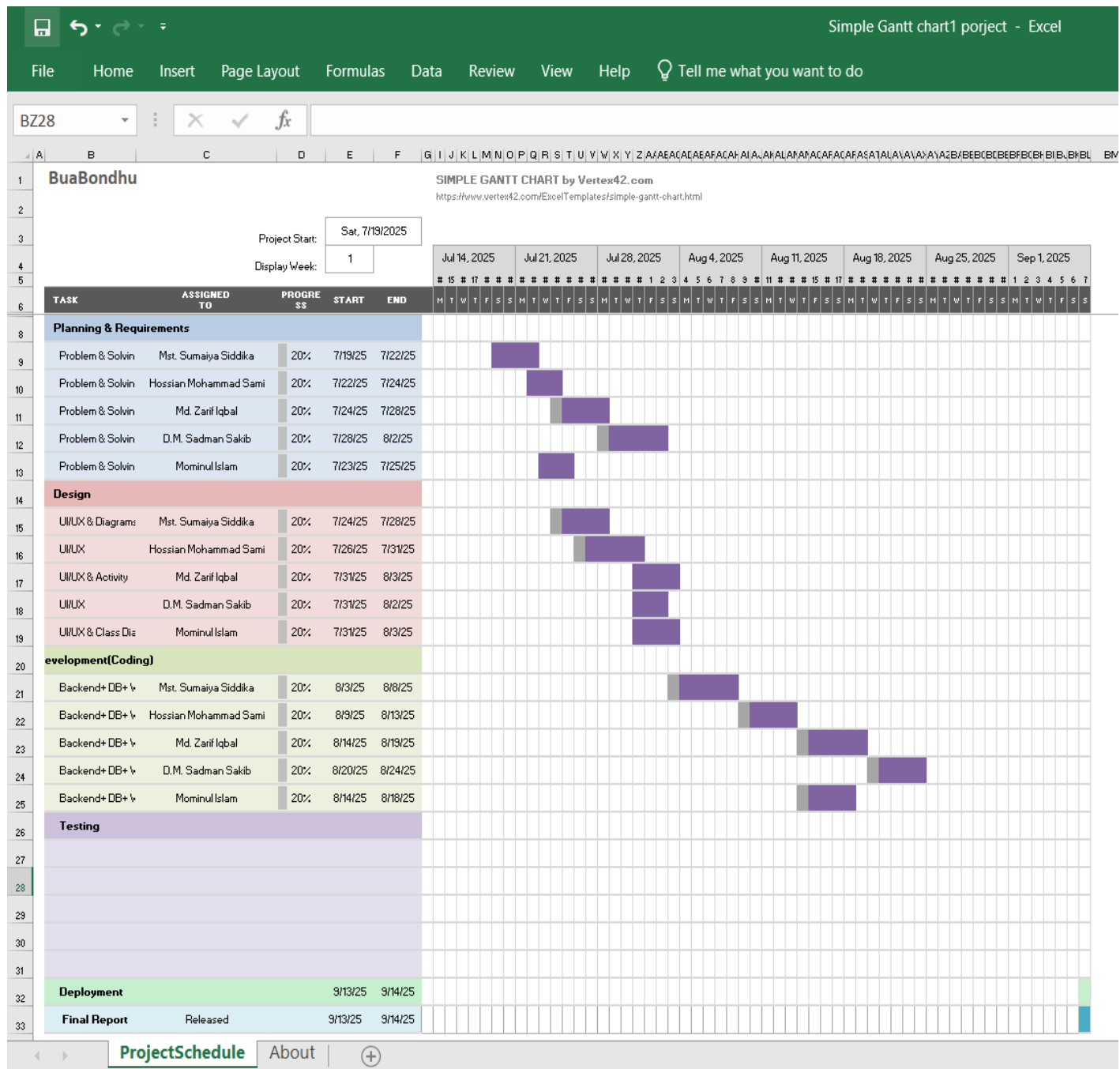
6.2 BUDGET

Item	Description	Estimated Cost (BDT)
App Development	Backend + Mobile + Web App	60,000
UI/UX Design	Layout, screens, testing	15,000
Database	Cloud server, storage	20,000
Security	OTP, encryption setup	10,000
Agency Training/Support	Helper onboarding cost	10,000
Marketing	Awareness campaign, posters, ads	15,000
Miscellaneous	Unexpected costs	5,000
Total		1,35,000 BDT

6.3 RISK

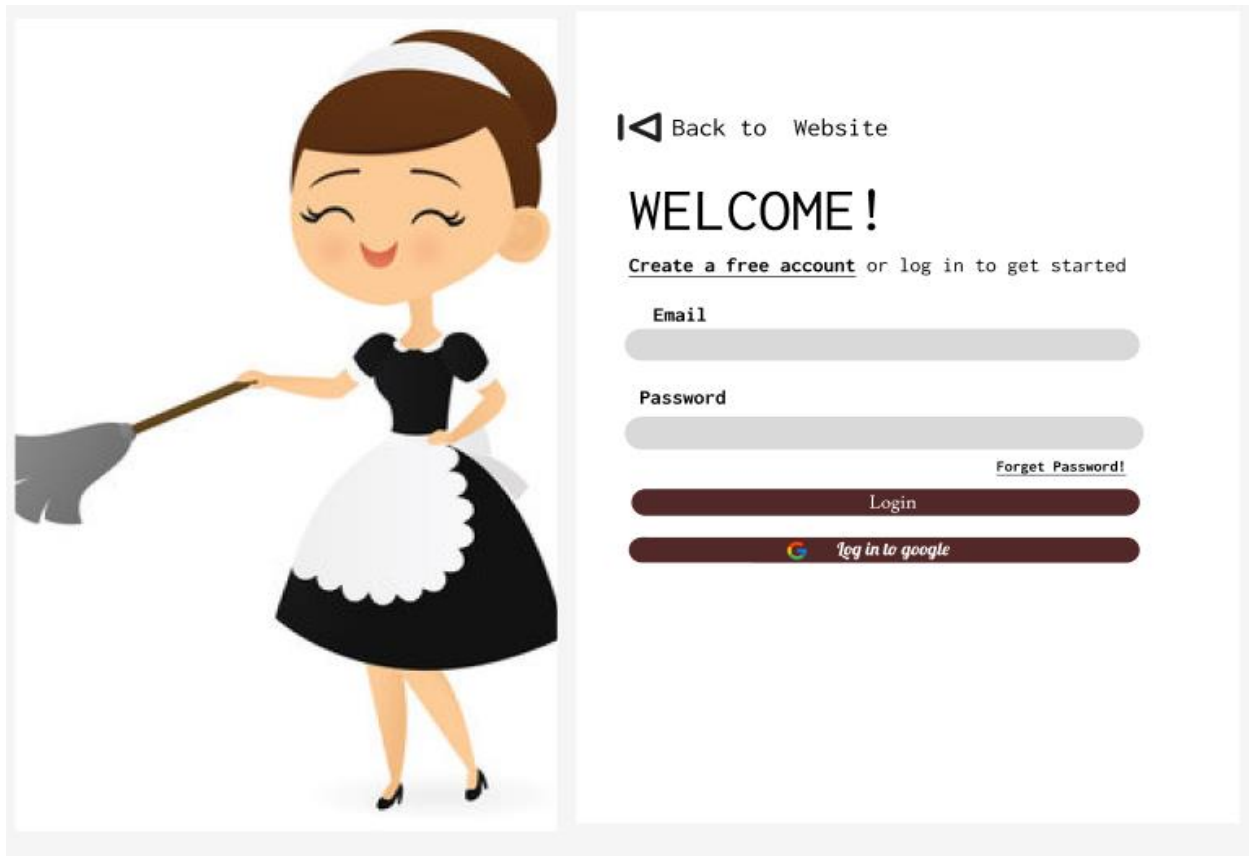
Risk ID	Risk Description	Risk Impact	Risk Response	Risk Level	Risk Owner	Notes
1	Helpers don't have NID or papers	Customer won't trust helpers	Agencies will check and accept other IDs if needed	High	Agency	Very important for trust
2	Data leak / hacking	Users lose trust, legal problems	Use encryption, OTP login, secure servers	High	Dev Team	Must keep data safe
3	Helpers can't use apps	Helpers can't register or update	Agencies will help them with profiles	Medium	Agency	Training may be needed
4	Helper cancels booking	Customer faces problems	Emergency replacement feature	Medium	System /Helper	Backup helpers needed
5	Server down	App won't work	Use cloud hosting and backups	High	Dev Team	Affects availability
6	Fake employer account	Helpers may face harassment	OTP login + ID check for employers	Medium	Dev Team	Prevent misuse
7	Miscommunication	Wrong job or disputes	Use in-app chat and booking records	Medium	System	Keep chat history
8	Few users join	Project may fail to grow	Awareness campaign + easy UI	Medium	Project Manager	Marketing important
9	Legal issues	Break law, face fines	Follow labor law, consult legal experts	Medium	Project Manager	Especially for payments
10	High cost	Hard to continue project	Start small, scale later, add subscription	Medium	Project Manager	Stay within budget

6.4 Scheduling (Gantt Chart)



7. Design

7.1 Prototype Design (UI/UX)



Frame 2

Create An Account

First Name	Last Name
Email Address	Phone Number
Choose a password	☐ I agree to Bua Bondhu's privacy policy and Terms of service.

Register

Frame 1



OTP Verification

Enter email and phone number to send one time password





Continue

Frame 2



Verification Code

We have sent the verification code to your email address

Continue

Main Users & Role

17:....

Customer

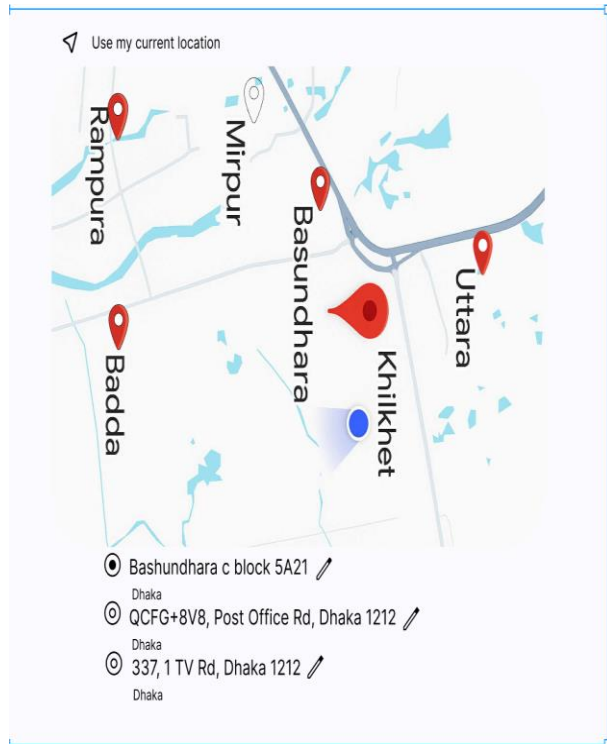
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Helper

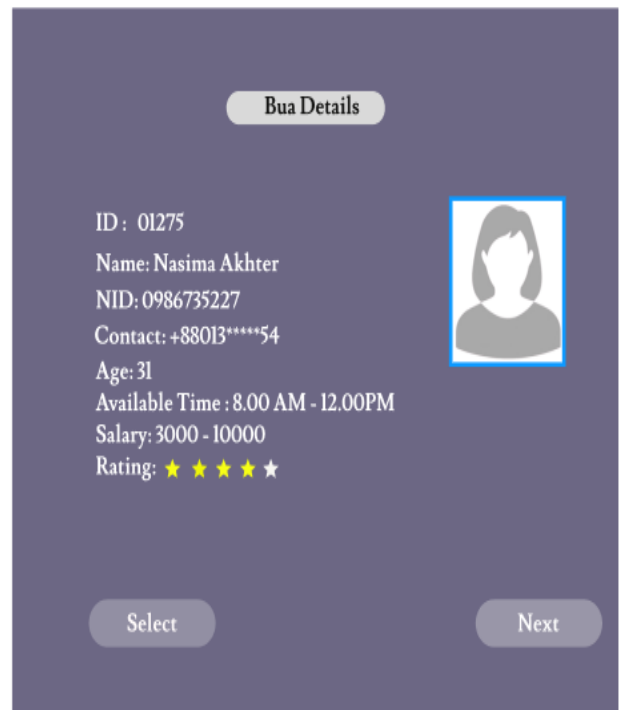
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Agency

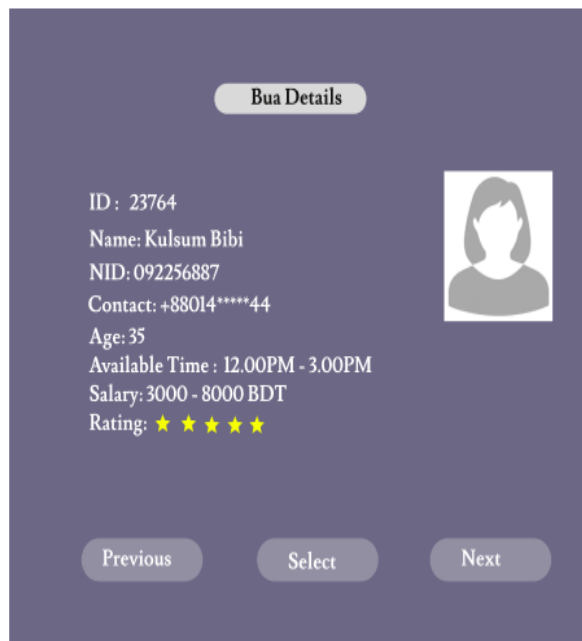

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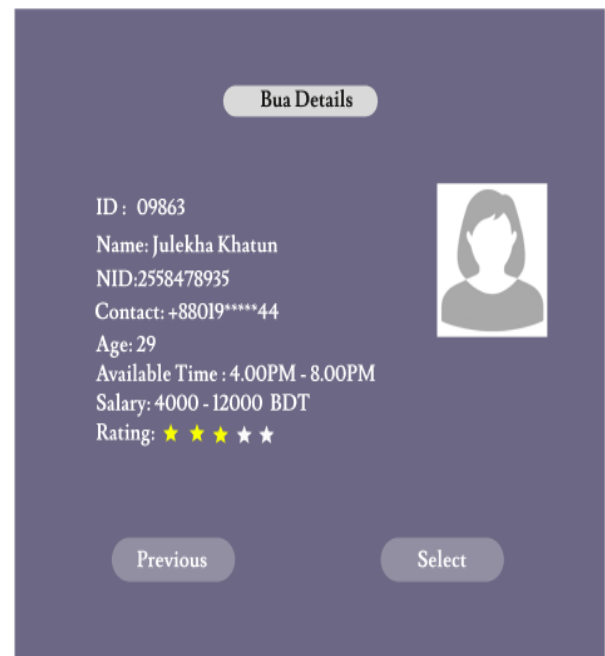
Frame 2



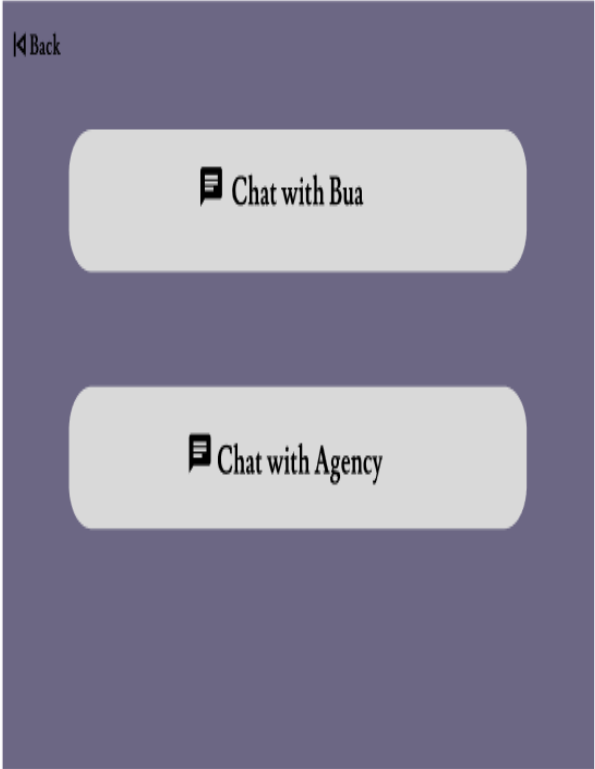
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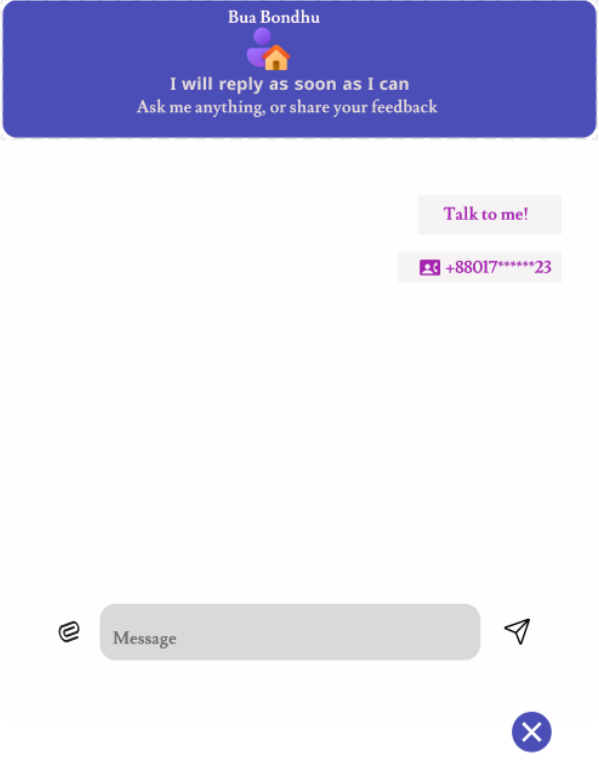
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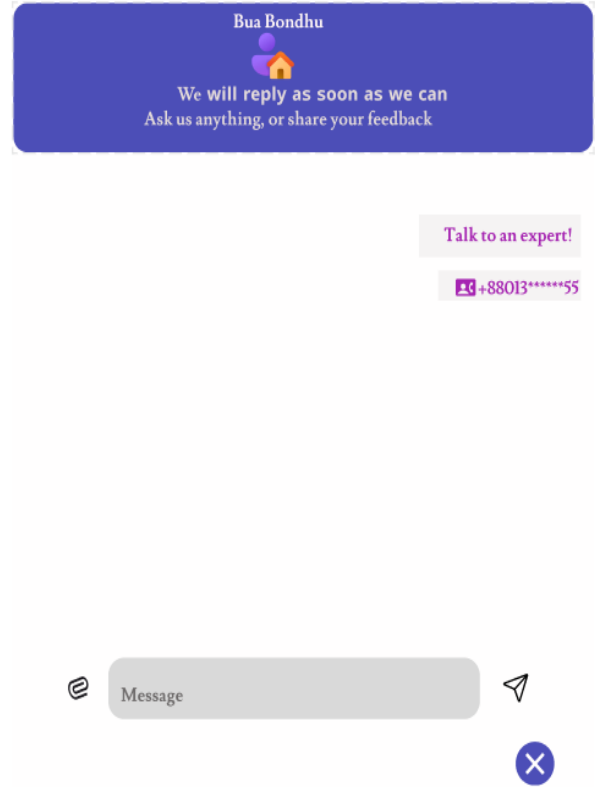
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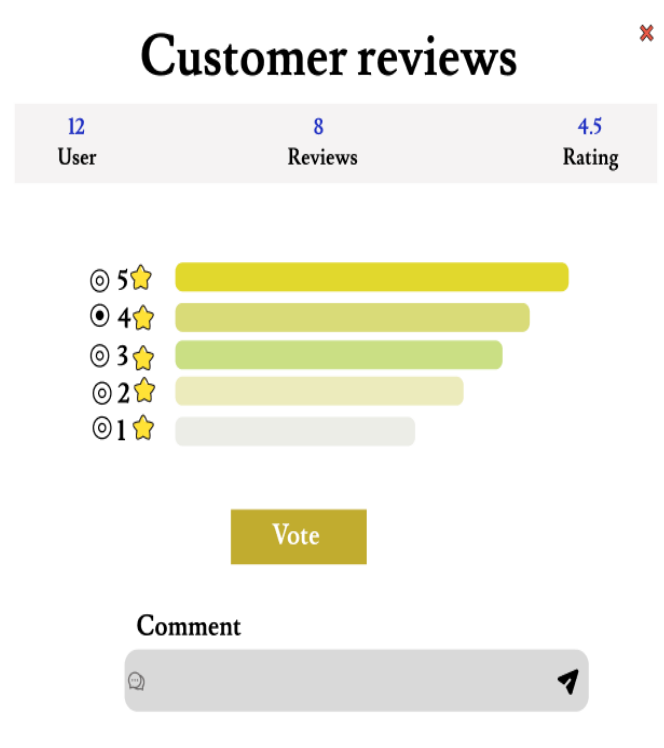
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Frame 1

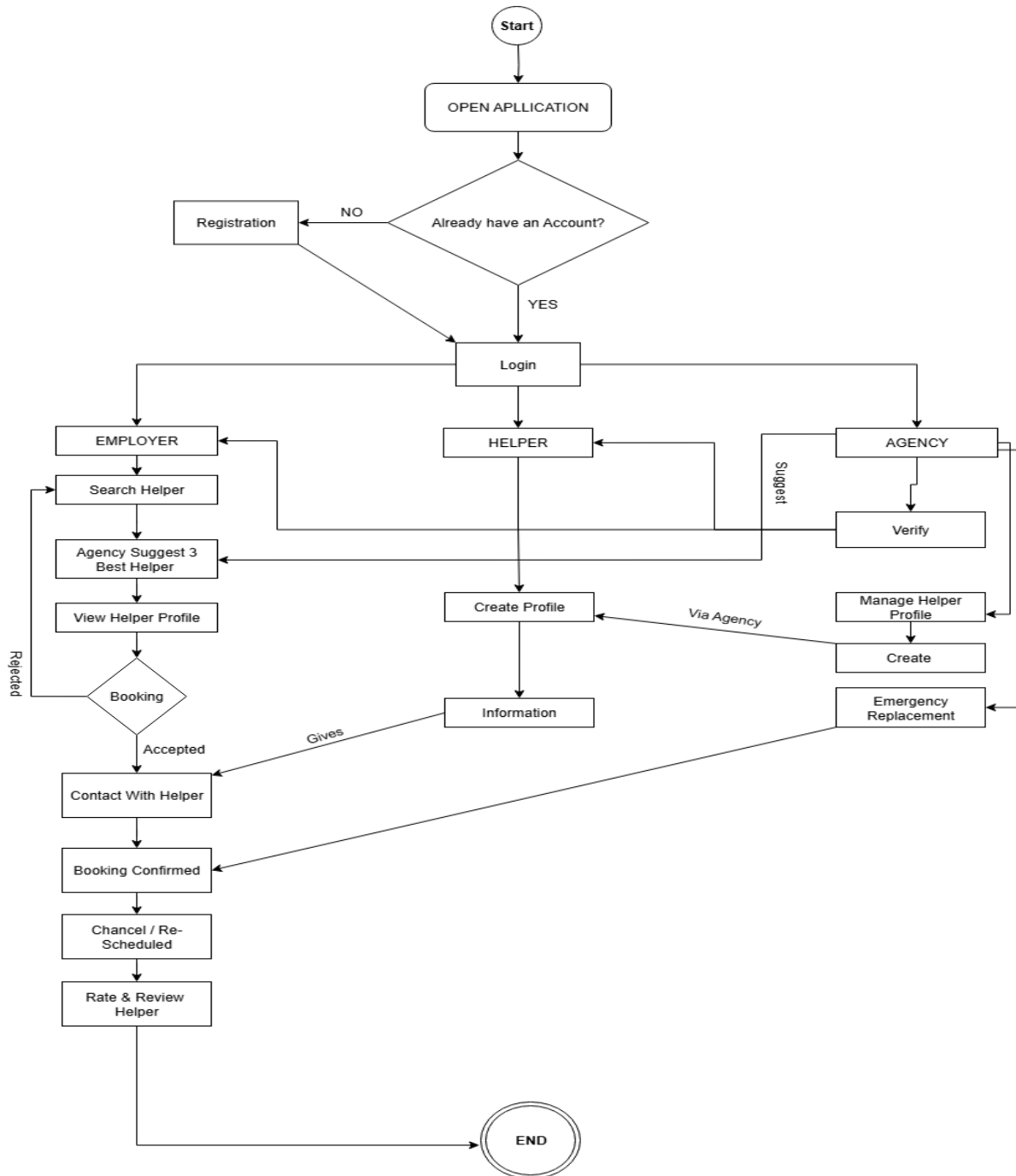


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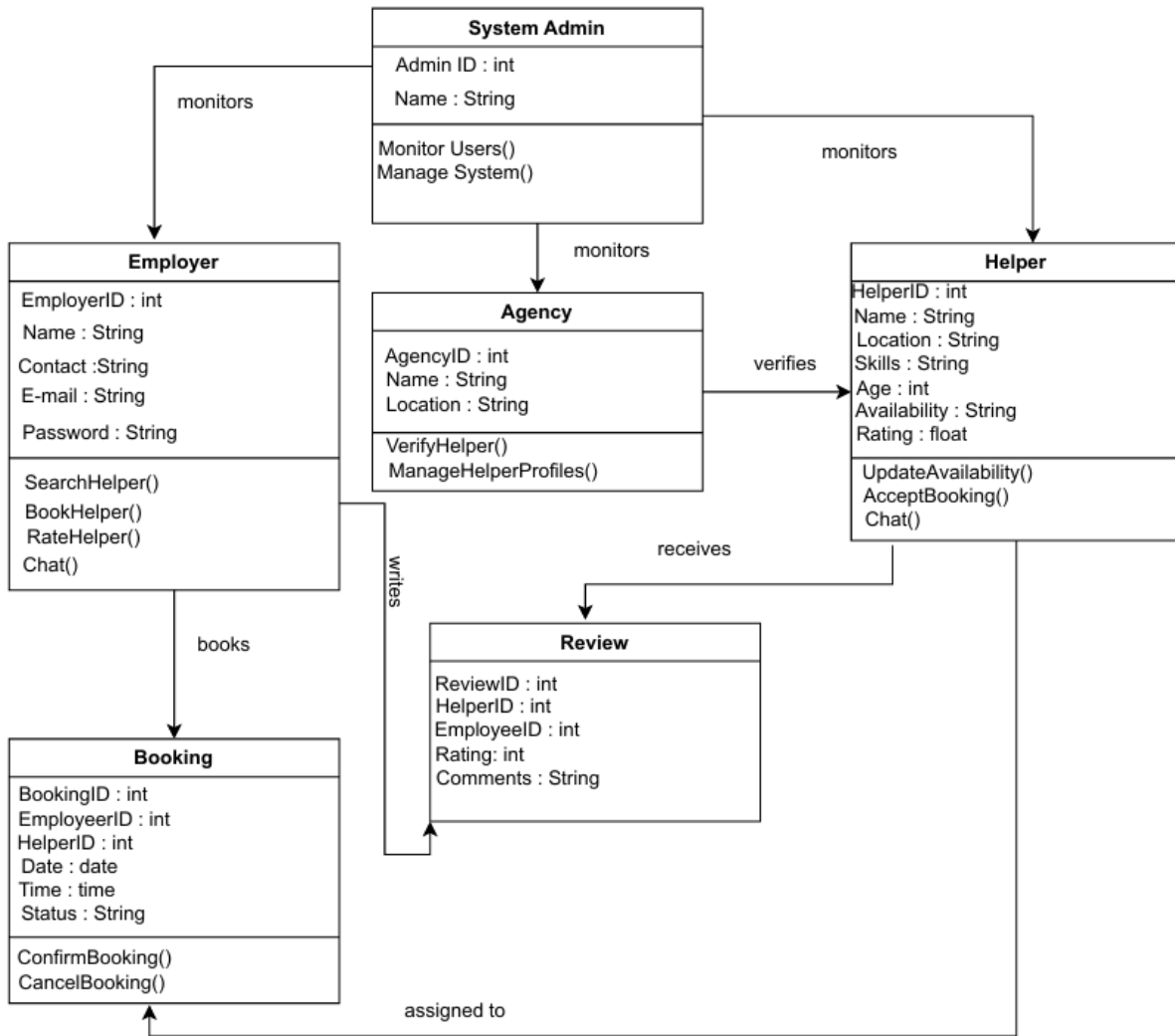


7.2 Diagram

a. Activity

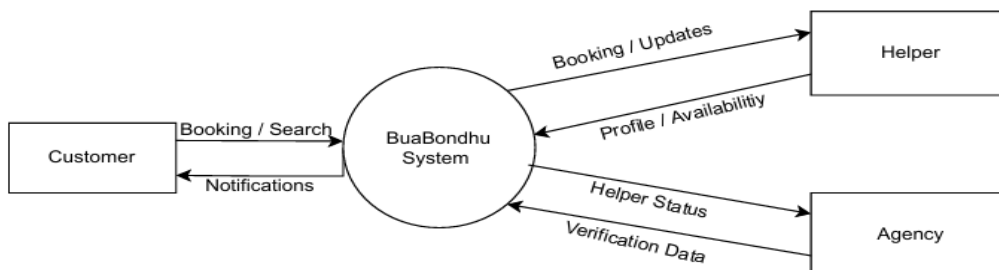


b. Class Diagram

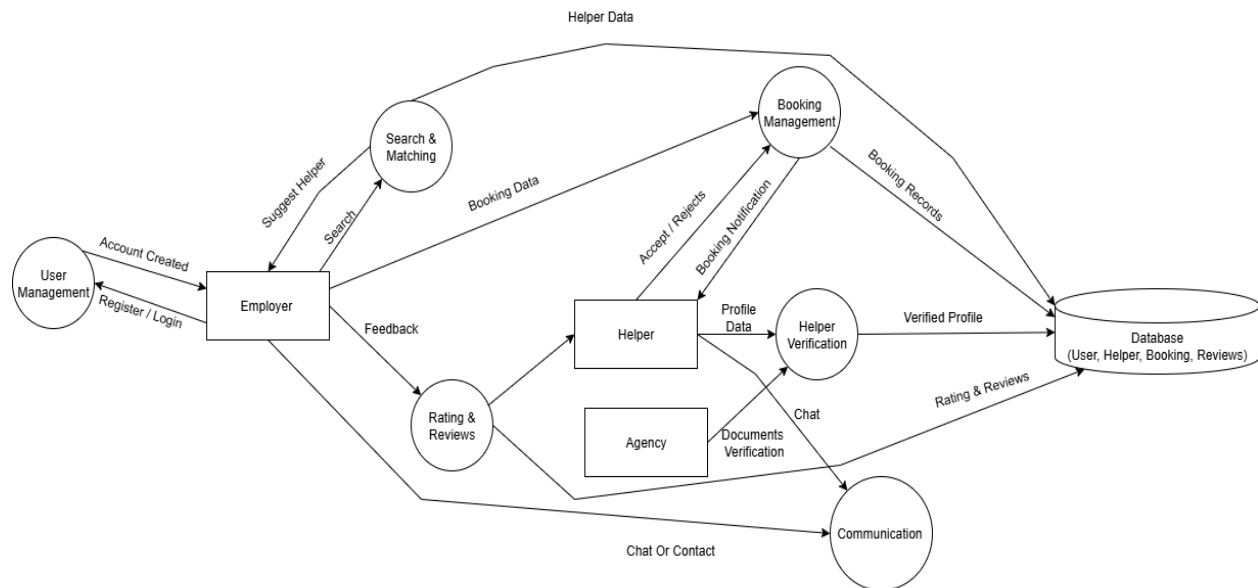


c. DFD Diagram

Level 0



Level 1



8. Conclusion

The **BuaBondhu Project** is made to help people easily connect with domestic helpers and agencies through a mobile and web app. In this project, important features like registration, booking, profile update, rating or reviews and security have been included. The work was done step by step – starting from planning, design, development, testing, and finally deployment. The system works as expected and is simple for users. This project shows how technology can solve daily life problems in an effective way.

Future Works

1. AI Integration
2. Online Payment
3. Multi-language App